Undavalli v surya prakash

Third Year Undergraduate
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Education

Degree	Institution	CPI/%	Year
B.Tech	SRM University	9.05	2023 – Present
Class XII	Tirumala Junior Kalasala	96.6	2021 – 2023

Projects

Calorie Burnt Prediction

- Built and deployed a predictive model using **XGBoost** to accurately forecast calories burnt with **97% prediction accuracy** based on activity level, duration, and individual biometric characteristics.
- Applied advanced machine learning algorithms to analyze **10,000+ real-world health and fitness records**, improving model generalization.
- Utilized statistical methods (**cross-validation, RMSE, MAE**) resulting in a **15% improvement** over baseline regression models.
- Reduced prediction error by **12%** and improved model training efficiency by **20%** through feature engineering and optimization.

Audio Deep Fake Prediction

- Developed a robust **Convolutional Neural Network (CNN)** to detect synthetic audio attacks, achieving an **AUC of 0.945** and **90% accuracy** on unseen data.
- Engineered a preprocessing pipeline converting raw audio into **Mel Spectrograms** and handled severe **9:1 dataset imbalance** using Random Up-sampling.
- Implemented **Spec Augment** (time and frequency masking) to mitigate overfitting and generalize across **13 unseen spoofing algorithms (A07A19)**.
- Achieved an **Equal Error Rate (EER) of 9.43%**, validating reliable detection of advanced voice clones.

Skill Summary

- **Languages:** Python, Verilog
- **Tools: MySQL, MATLAB, Simulink, PSpice**
- **Web Technologies: HTML, CSS, JavaScript**
- **Frameworks: Spring Boot, React**
- **Libraries: NumPy, Pandas, Matplotlib, Scikit-learn**
- **Coursework: Data Structures & Algorithms (DSA), OOPS, SQL**
- **Areas of Interest: Machine Learning, Data Analysis**
- **Soft Skills: Problem Solving, Project Management, Team Collaboration, Self-learning, Presentation**

Extra-Curricular Activities

- Cricket & Badminton